```
% Clear workspace and figures
clear; close all; clc;

% Filter specifications matching the output image[cite: 7]
N = 40;              % Filter order
fs = 10000;          % Sampling frequency (Hz), Nyquist is 5000 Hz
f_cutoff = 1800;     % Cutoff frequency for transition band

% 1. Parks-McClellan (Equiripple) Filter Design
% Frequency vector normalized from 0 to 1 (where 1 corresponds to Nyquist
fs/2)
f_spec = [0, f_cutoff, 2200, fs/2] / (fs/2);
a_spec = [1, 1, 0, 0];
b_pm = firpm(N, f_spec, a_spec);

% 2. Frequency-Sampling Filter Design (using fir2 interpolation)
f_fsamp = [0, f_cutoff/(fs/2), 2200/(fs/2), 1];
a_fsamp = [1, 1, 0, 0];
b_fs = fir2(N, f_fsamp, a_fsamp);

% Compute Frequency Responses
[H_pm, f] = freqz(b_pm, 1, 2048, fs);
[H_fs, ~] = freqz(b_fs, 1, 2048, fs);

% Convert Magnitude to dB
mag_pm_db = 20 * log10(abs(H_pm) +eps); % Adding eps to avoid log(0)
mag_fs_db = 20 * log10(abs(H_fs) + eps);

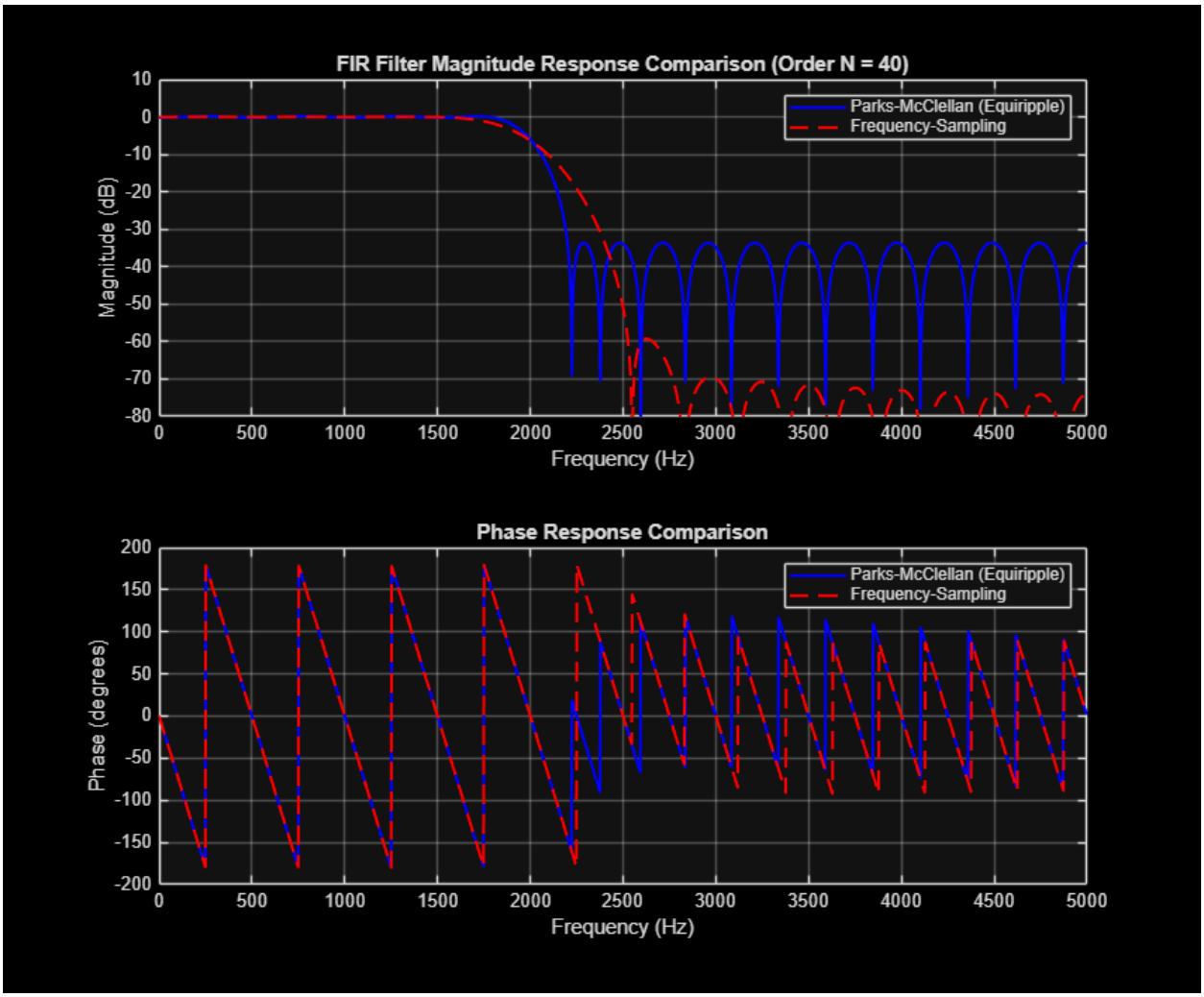
% Convert Phase to Degrees (-180 to 180 range)
phase_pm = angle(H_pm) * (180 / pi);
phase_fs = angle(H_fs) * (180 / pi);

% --- Plotting Results ---
figure('Name', 'Figure 1: Parks-McClellan vs Frequency-Sampling', 'Position',
[100, 100, 850, 700]);

% Subplot 1: Magnitude Response Comparison
subplot(2,1,1);
plot(f, mag_pm_db, 'b-', 'LineWidth', 1.5);
hold on;
plot(f, mag_fs_db, 'r--', 'LineWidth', 1.5);
title('FIR Filter Magnitude Response Comparison (Order N = 40)');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
legend('Parks-McClellan (Equiripple)', 'Frequency-Sampling', 'Location',
'northeast');
grid on;
xlim([0, fs/2]);
ylim([-80, 10]);

% Subplot 2: Phase Response Comparison
subplot(2,1,2);
```

```
plot(f, phase_pm, 'b-', 'LineWidth', 1.5);
hold on;
plot(f, phase_fs, 'r--', 'LineWidth', 1.5);
title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
legend('Parks-McClellan (Equiripple)', 'Frequency-Sampling', 'Location',
'northeast');
grid on;
xlim([0, fs/2]);
ylim([-200, 200]);
```



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